ADO SNOWGRID CL

Validation report

2021-12-14; ZAMG; Konrad Mayer, Klaus Haslinger

Snow model

ADO Datasets from deliverable D.T2.1.1 are downscaled from ERA5 surface datasets (Hersbach et al. 2018) using quantile mapping. As ERA5 snow data cannot be downscaled in a similar manner (no snow data available from UERRA mescan surfex; Bazile et al. 2017) and its accuracy within the alpine domain was questioned, a modified version of the deterministic snow model SNOWGRID-CL (Olefs et al. 2020) was used to derive daily fields of snow depth and snow water equivalent within the scope of the ADO project (part of deliverable D.T1.2.1).

Design changes, deviating from the published version, include:

- use of wet bulb temperature as a proxy of snow temperature
- use of wet bulb temperature and a sigmoid function to derive solid fraction of the precipitation. Parametrization as published by Wang 2019 was used.
- use of actual solar radiation (direct and diffuse)
- no use of lateral snow redistribution mode (processes are not resolved at 5.5km spatial resolution)
- no use of undercatch correction for all elevations

Additionally, several technical improvements and adaptions, e.g. porting to python 3, adapted data import and output, unit testing, logging etc. were implemented in the ADO version of the model.

In order to run the model, the meteorological datasets from deliverable D.T2.1.1 needed to be extended by several additional input datasets and static fields, described in more detail in the following sections.

Static Albedo

The method of Brock et al. 2000 is used in SNOWGRID-CL to calculate the actual albedo from a static field of snow free albedo, the snow depth as well as accumulated maximum temperature. Static albedo was derived from corine land cover data (CLC 2018) as follows:

- reprojected from EPSG 3035 to UERRA Projection (Lambert Conformal Conic projection, refer to the UERRA documentation for details on the exact parameters or proj4 string).
- Clipped to ADO bounding box
- lookup operation for reference albedo values per land cover class (from literature)
- upsample to ADO grid (100m to 5.5km) as a weighted mean of subpixel albedo values

Potentially too high values appear in grid cells with a high share of CORINE land cover class 331 *Beaches dunes sands* (e.g. in Croatia or at Šumava National Park/Nationalpark Bayerischer Wald). However, these do not appear within the EUSALP domain (see figure 1).

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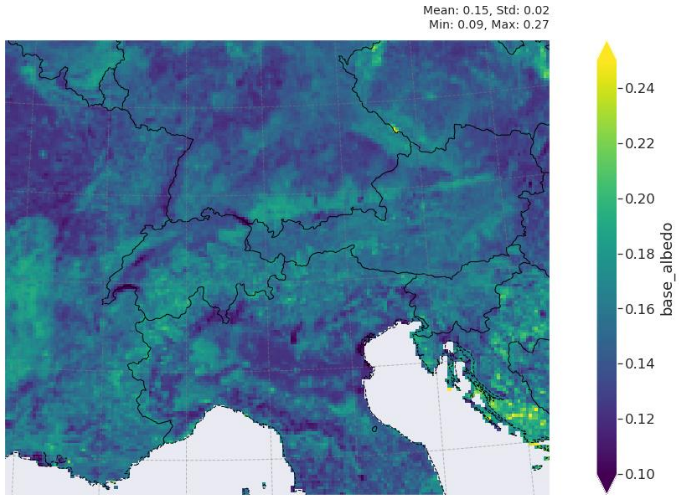

Figure 1:

static snow-free albedo derived from CORINE land cover classes.

Radiation

SNOWGRID-CL implements the enhanced temperature index method to model snow ablation (Pellicciotti et al. 2005), considering actual albedo and global radiation in addition to surface temperature. As ERA5 data is provided on the horizontal surface, the direct radiation component needs to be corrected for grid cell inclination using the UERRA digital elevation model which is done in hourly resolution (after interpolating UERRA data, which is only available in 6 hour aggregates) and aggregated to daily values after correction. This procedure is applied to data of direct solar radiation before being used as input data to the snow model. As can be seen in Figure 2, the correction complies with the pronounced topography in the alpine region and attributes more incoming energy to southwards exposed slopes. However, shading effects are not accounted by this correction.

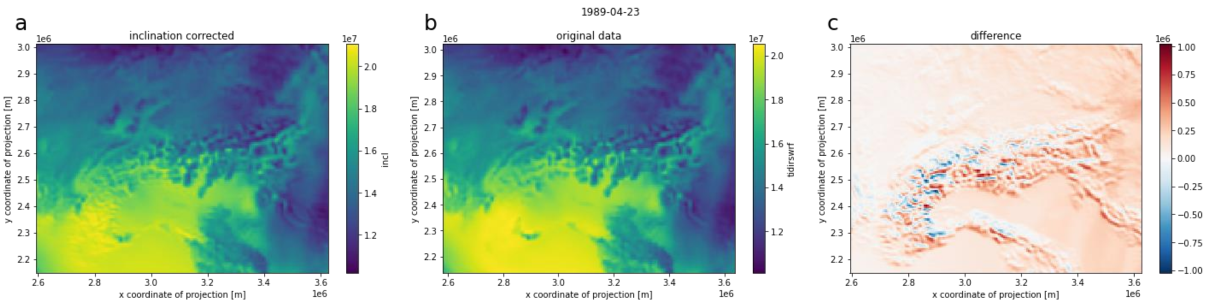

Figure 2: Inclination correction of direct solar radiation (in J/m² resp. Ws/m²); a) inclination corrected data, b) original downscaled data on the horizontal surface, c) difference (corrected-original).

Digital elevation model, slope, aspect

The digital elevation model as provided with the UERRA MESCAN SURFEX dataset, at a spatial resolution of 5.5km, was used. Derived variables, slope and aspect, were calculated with the package xarray-spatial (https://xarray-spatial.org/).

Maximum Temperature

The method of Brock et al. 2000 is used in SNOWGRID-CL to calculate the actual albedo from a static field of snow free albedo, the snow depth as well as accumulated maximum temperature. Maximum temperature can be derived

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from hourly ERA5 data, but UERRA data needed for downscaling is not available. Temperature varies on large spatial scales, but is strongly mediated by orography. We therefore chose a geostatistical approach, which takes surface elevation as external drift into account (external drift kriging with a Gaussian kernel). As this method is computationally very expensive the data was decomposed into a day-of-year-climatology, the mean maximum temperature per day of the year (DOY), and the respective daily anomaly. External drift kriging was performed on the climatology only, whereas the anomaly was bilinear interpolated. The elevation signal is expected to be represented within the climatology only. Therefore, the much faster bilinear interpolation of the anomalies varying on large scales in space is reasonable and the computationally expensive Kriging limited to an a priory calculation of 366 fields. Summed interpolated climatology and interpolated daily anomaly represent the respective interpolated output (see fig 3).

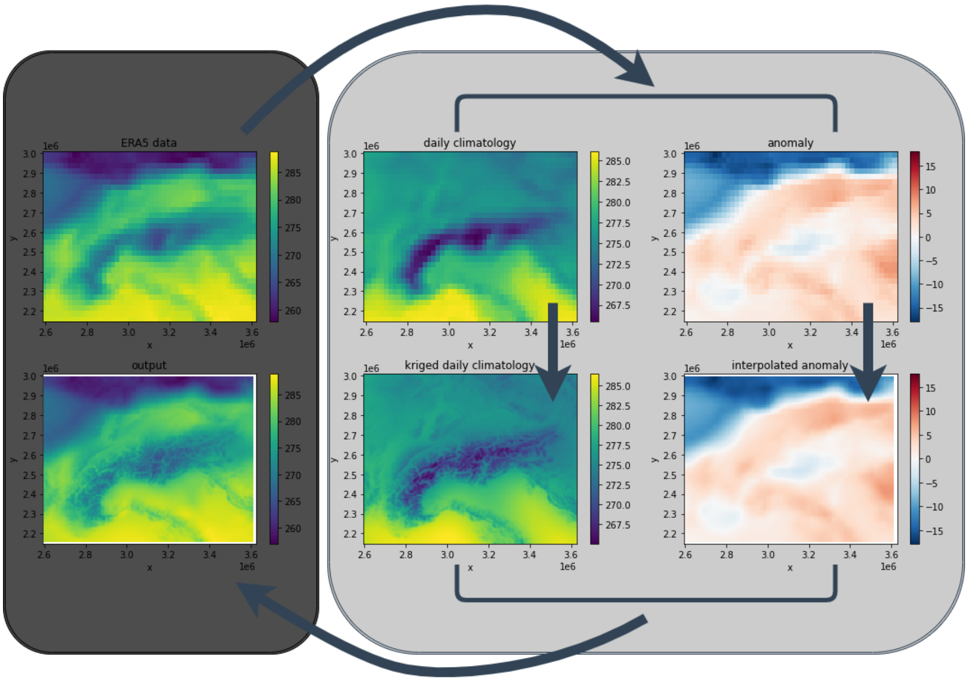

Figure 3:

Spatial interpolation of maximum temperature using external drift kriging of the DOY-climatology and a superimposed binlinear interpolated anomaly.

Reference dataset

A comprehensive dataset of observed snow depth is available through the project CliRsnow (https://clirsnow.netlify.app/) for the entire alpine region. Most of the station data is available via Creative Commons Attribution 4.0 International Public License on a public data repository (https://zenodo.org/record/5109574#.YbHNj3VKhhE, Matiu et al. 2021). Separate declarations of consent were obtained from data providers not covered by the license stated above, for the extent of the validation of the snow model output within the project ADO.

A total of 2950 stations are available in the CliRsnow dataset, of which 22 were excluded from the validation dataset as they fall onto grid cells known to show unreliable model output (red points in fig 4a). Of the remaining stations, skill scores were only calculated for overlaps with the model output of more than ten years (a condition met at 2694 stations).

The ADO Datasets from deliverable D.T2.1.1 used as input data to the deterministic snow model SNOWGRID-CL are downscaled data with a spatial resolution of 5.5km (from 31km). Thus, scale problems are evident when comparing model output to the observed station data. Figure 4b shows the relationship of the station elevation with the associated SNOWGRID grid cell elevation. Deviations are partly substantial, especially for towards the higher elevations, but the linear relationship is perceptible.

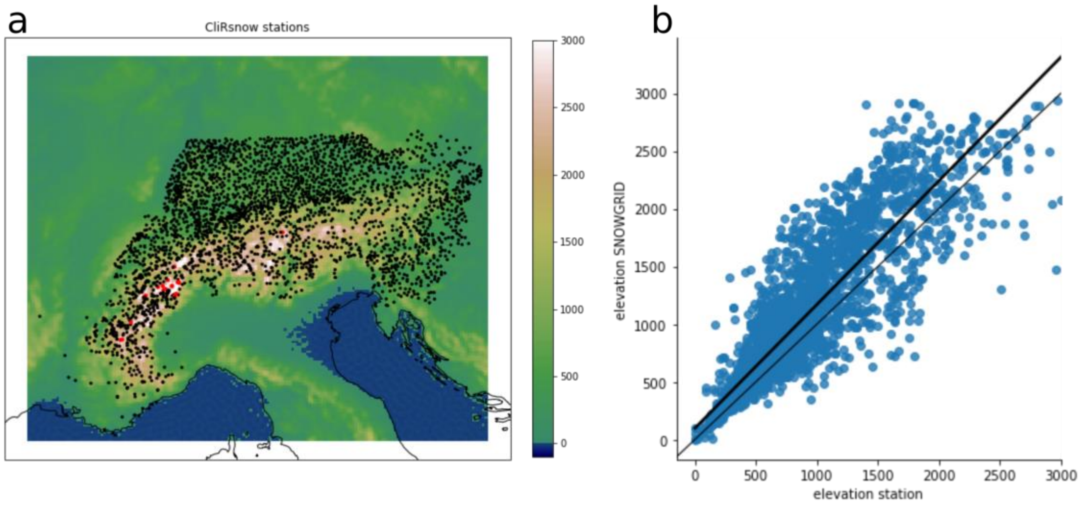

Figure 4:

Validation data from CliRsnow;

a) spatial distribution of stations (masked stations in red),

b) station elevation vs. elevation of the ADO input data (UERRA elevation).

Validation results

Glacier mask

The coarse spatial resolution of the ADO version of SNOWGRID-CL impedes the use of a lateral snow redistribution scheme as gravitational processes happen on much smaller scales. Even when considering gravitational redistribution, such as done by Olefs et al. 2020, the simple parametrization scheme without considering wind data cannot prevent the formation of snow towers. Also, SNOWGRID-CL does not implement processes within glaciers and thus does not output correct snow depth and snow water equivalent time series at such locations.

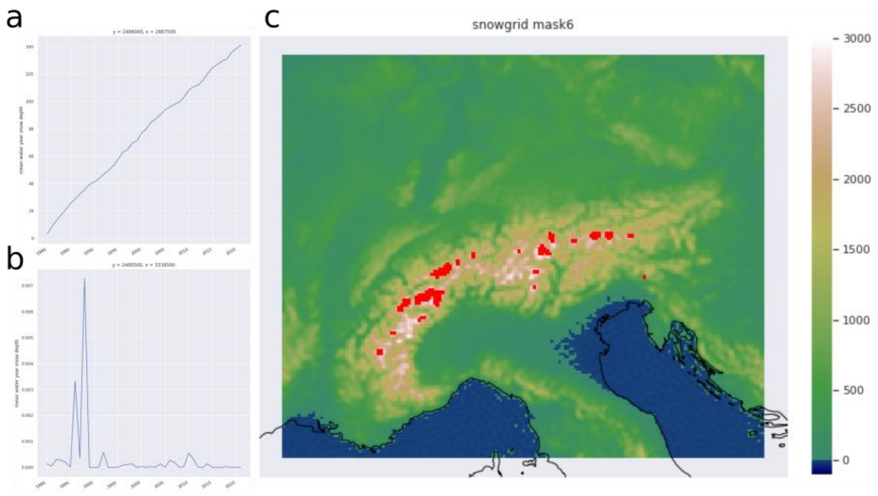

Figure 5:

glacier mask derived using series autocorrelation;

a) highly autocorrelated grid cell time series of mean water year snow depth,

b) grid cell time series with low autocorrelation,

c) masked grid cells.

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For the analysis within the scope of this validation report, grid cells with very strong autocorrelation within the time series of mean snow depth per water year were masked as they are representing grid cells coinciding with glacier area (see fig 5c, panels a and b give examples for a highly autocorrelated and a non-autocorrelated grid cells). This was done using Ljung-Box Q Test (Ljung and Box, 1978), excluding all grid cells with a p-value smaller than 0.00001 at lag 1. It is adviced to **exclude these grid cells from further downstream analysis** (e.g. the calculation of snowpack index, SSPI) within ADO (use **mask6** within snowgrid_masks.nc for this purpose).

Skill assessment: snow depth

Daily differences in snow depth

RMSE was calculated on daily basis, showing a mean value of 21cm and a maximum value of 232.4cm. The mean stationwise mean difference was -5.5cm, highest absolute difference (-)186.8cm. However, given the high spatial variability mediated by covariates (mainly indirectly and directly driven by orography), global statistics are of limited relevance as they are strongly influenced by the specific locations of the stations in the reference dataset. More information is contained in the patterns of spatial variability of the different skill scores. Higher values of RMSE and mean difference can be observed within the alpine arc, whereas low values cluster in the lower elevations north and east of the Alps as well as in bigger valleys (see figure 2).

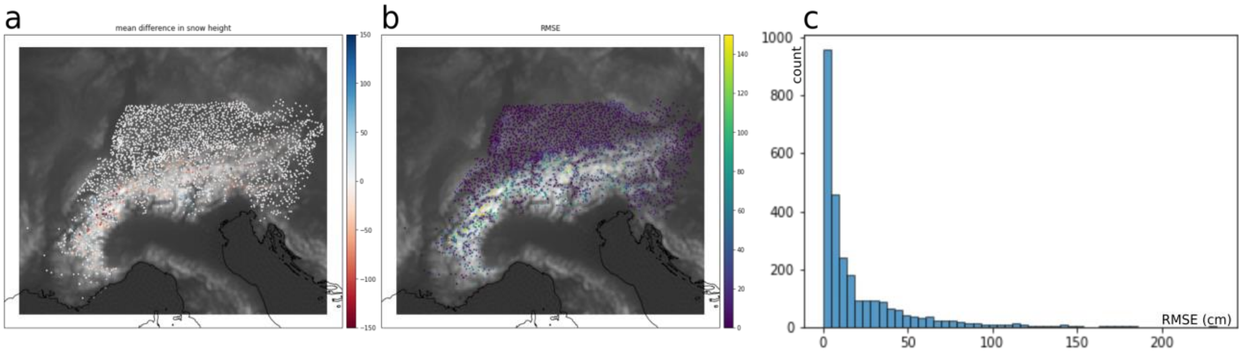

Figure 6: model skills from daily snow depth values; c) Histogram of RMSE, b) spatial distribution of RMSE, s) spatial distrubution of mean differences (blue-observations show higher values, red-model shows higher values). The color scales in (a) and (b) are truncated.

These high values can be attributed to different drivers such as the elevation of the grid cell, the magnitude of the variable (modeled snow depth) or the difference in elevation of the observation station and the model grid cell, with the strongest (linear) relationship tied to the magnitude of snow depth (higher error, the higher the modeled snow depth, see fig 2), and weaker non-linear relationships for the grid cell elevation (with high spread >800m) and the absolute elevation difference (elevation delta, please refer to section "" for more details on the latter).

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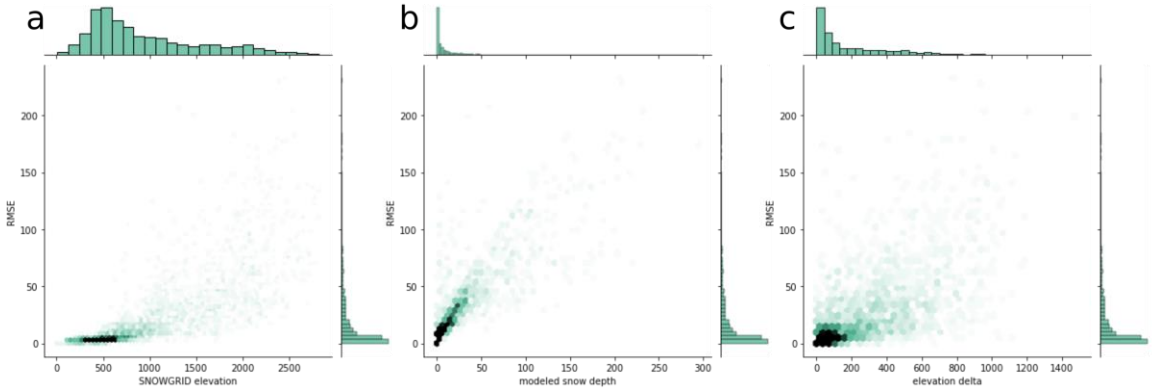

Figure 7: Potential sources of error; a) error increases strongly from >800 meters with hughe spread (nonlinear), b) error is mainly linked to the magnitude of modeled snow depth (linear), c) weakly tied to the deviation in elevation between station and grid cell.

Annual differences in snow depth

Similar spatial patterns can be observed for mean water year snow depth (see fig 14), but errors are slightly lower at this temporal aggregation (mean RMSE 13.5, max RMSE 196.2, mean water year mean difference -5.3, max absolute mean difference (-)186). Again, a distinct link of error magnitude to grid cell elevation can be seen (fig 14b), with higher errors at higher elevations and a slight tendency to a negative mean difference in snow depth, meaning a more often occurring overestimation of snow depth in the model output, which can also be observed in the histogram (fig 14a, fig 14b).

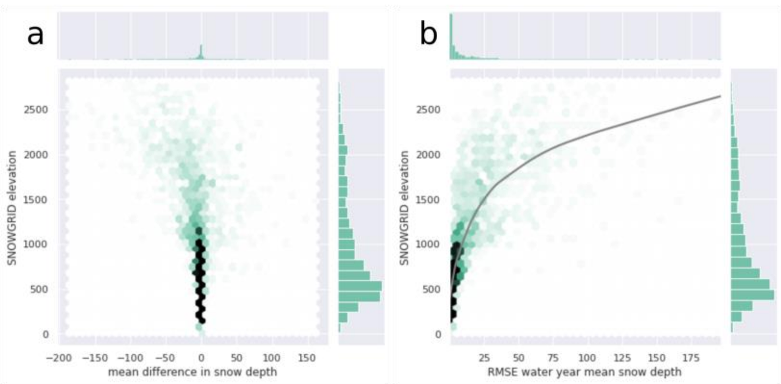

Figure 8: relationship of SNOWGRID grid cell elevation with a) mean difference of snow depth, b) RMSE (with LOWESS smoother as gray line)

Seasonal differences in snow depth

From figure 15, we can deduce that highest errors appear in early Summer (April to June). However, as these high errors only appear in a handful of stations, this is hardly seen in global statistics or histograms (not shown), which are dominated by the many low elevation stations. These are already snow free in this season and thus force the mean RMSE to lower values or dominate the histogram at 0 or very low values. By excluding stations with observation and model output concurrently being 0, a distribution with a mode at even lower values for AMJ as compared to JFM, but showing some very high values, can be observed (not shown).

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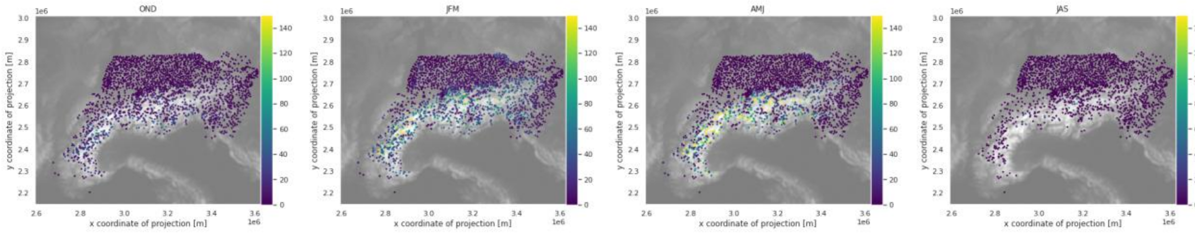

Figure 9: RMSE for mean snow depth in different seasons of the water year. OND) October to December, JFM) January to March, AMJ) April to June, JAS) July to September

Although highest errors appear in April to June (AMJ), they do not necessarily origin from a misrepresentation of the ablation phase in SNOWGRID-CL. Also, an outsized snow accumulation of snow leads to an overlong snow cover duration and higher errors in the ablation phase.

Skill assessment: snow cover duration

Snow cover duration (SCD) was defined as the number of days within a water year with a snow depth greater than 1cm. On average, the model overestimated SCD by more than two weeks (15.8 days, see distributions in fig 15). In many cases SCD observations were reproduced realistically (fig 10a). In cases with higher errors, the model reproduced the inter-annual variability still quite well, but at an offset (fig 10b). As raster data at a relatively coarse resolution are compared with point (station) data, scale problems are evident. The error is a function of the difference in elevation between observation and model grid cell (fig 16). Therefore, the differences in scale may explain observed offsets.

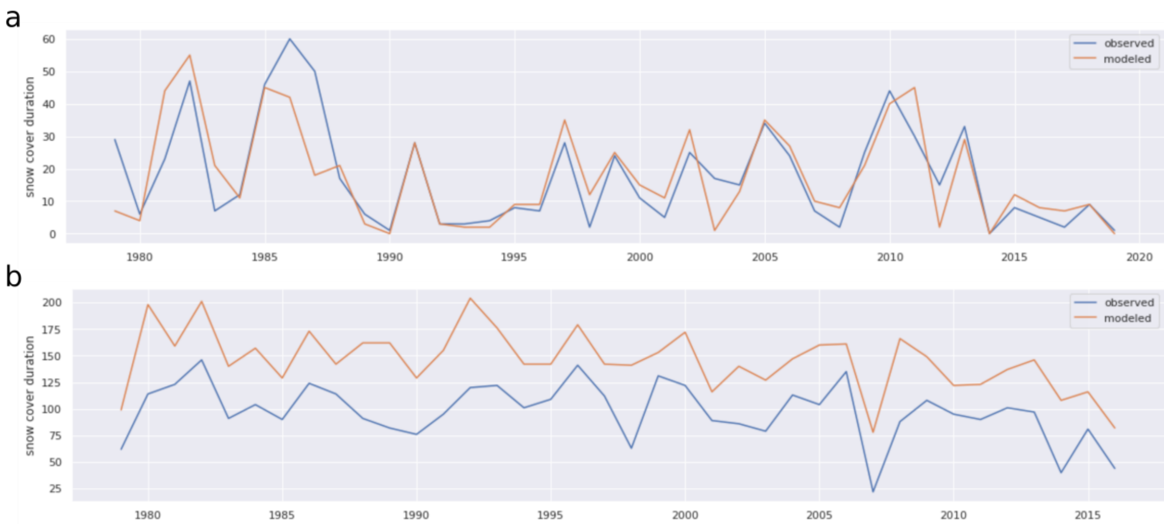

Figure 10: Snow cover duration: a) reasonable model output at the station Markgroningen, b) offset, but realistic interannual variability at station Admont

Potential sources of error/offset

Observed scale problems in the preceding section are also evident for model output of snow depth, but deviations of the model output from the observed values may also originate from processes not implemented in the model or biased input data. The following sections briefly discuss these potential sources of error.

Input data

The modeled snow data is highly dependent on the input data from the ADO dataset which is downscaled from ERA5 reanalysis data. ERA5 tends to overestimate precipitation (Shrestha et al. 2021), especially in winter (Keller 2021). Observed overestimation (on average) of snow depth (fig 14b) and snow cover duration (fig 15b) may therefore just represent a wet bias in the input data.

Scale problems

Model output represent grid cells with 5.5km spatial resolution, station data represent single points in space. A grid cell therefore aggregates a much bigger area, which may not resolve the topography in mountainous areas sufficiently. Thus, the elevation of a station can deviate considerably from the elevation of the associated grid cell (see fig 4b), especially in higher elevations (fig 17). The influence of this deviation is not too obvious looking at RMSE and the elevation delta (which is the absolute elevation difference, see fig 7c). Considering not only the magnitude, but also the sign of both measures reveals their nonlinear relationship (fig 11b and c). Small elevation differences are only weakly reflected in differences in snow depth. In case the station is located at considerably higher (lower) elevation, i.e. the elevation difference is positive (negative), the model outputs a lower (higher) magnitude of snow depth, i.e. a positive (negative) mean difference in snow depth (fig 11c).

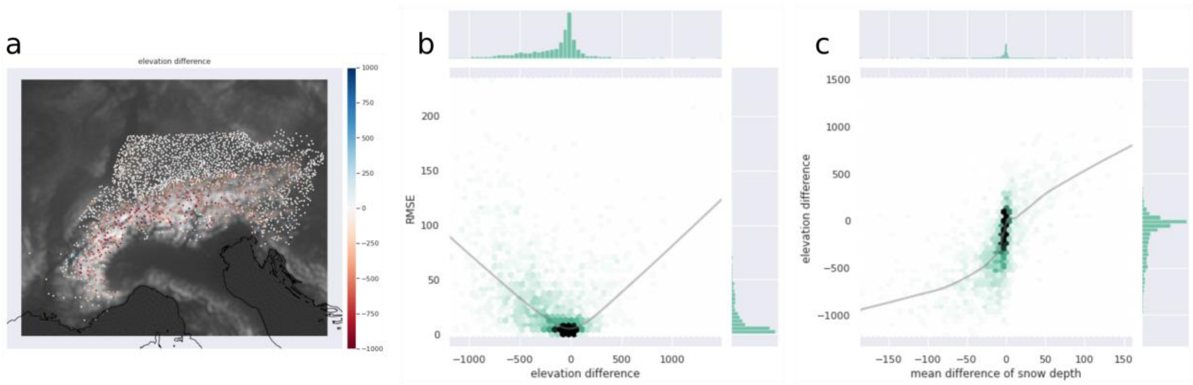

Figure 11: a) Difference in elevation between station and SNOWGRID grid cell (negative, red if station is lower; positive, blue if station is higher); b) RMSE gets higher at higher differences in elevation; c) negative difference (station lower) is associated with a negative difference in snow depth (higher modeled values), positive elevation differences with positive differences in snow depth (lower modeled values). Gray lines represent LOWESS smothers.

This can be also observed looking at time series at specific stations and their associated grid cell time series. At “Zugspitze”, the station showing the biggest elevation delta, the grid cell elevation is 1487.6m lower than the station elevation (station at 2964m, grid cell at 1476.4m). The station is obviously not representative for the grid cell area and much higher snow depths are observed relative to the modeled snow depths (fig 14a). The opposite case is observed at the station “Ginzling“, located 1124 below the associated grid cell elevation (fig 14b, station at 1060m, grid cell at 2184m).

At stations with very low elevation delta, no such offsets of snow depth magnitude can be observed (fig 13).

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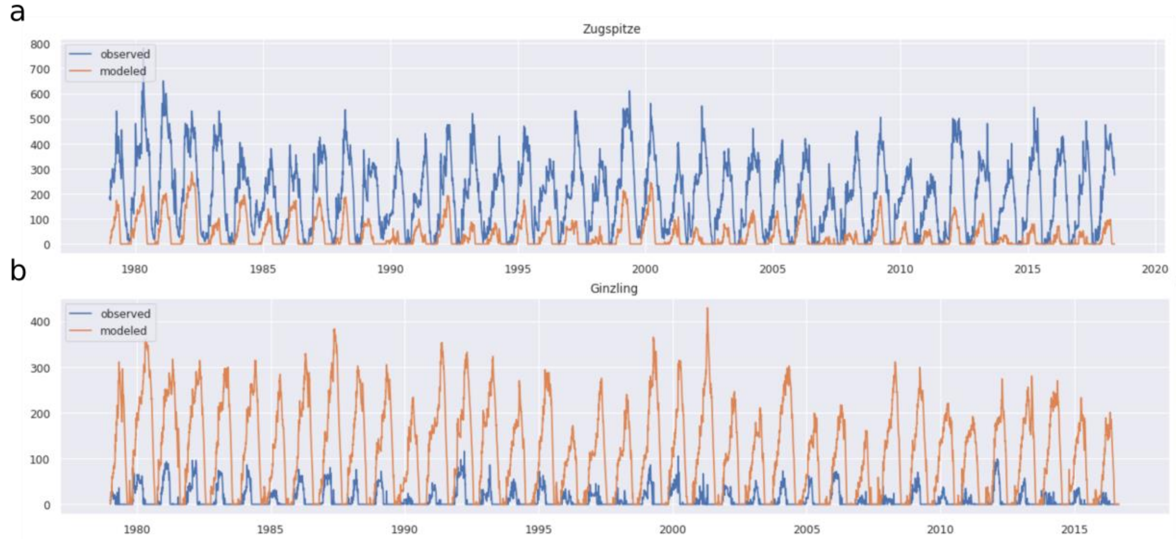

Figure 12: Stations with high differences in elevation relative to the SNOWGRID grid cell elevation; a) station Zugspitze is located 1487.6m above the grid cell elevation, b) station Ginzling is located 1124 below the grid cell elevation.

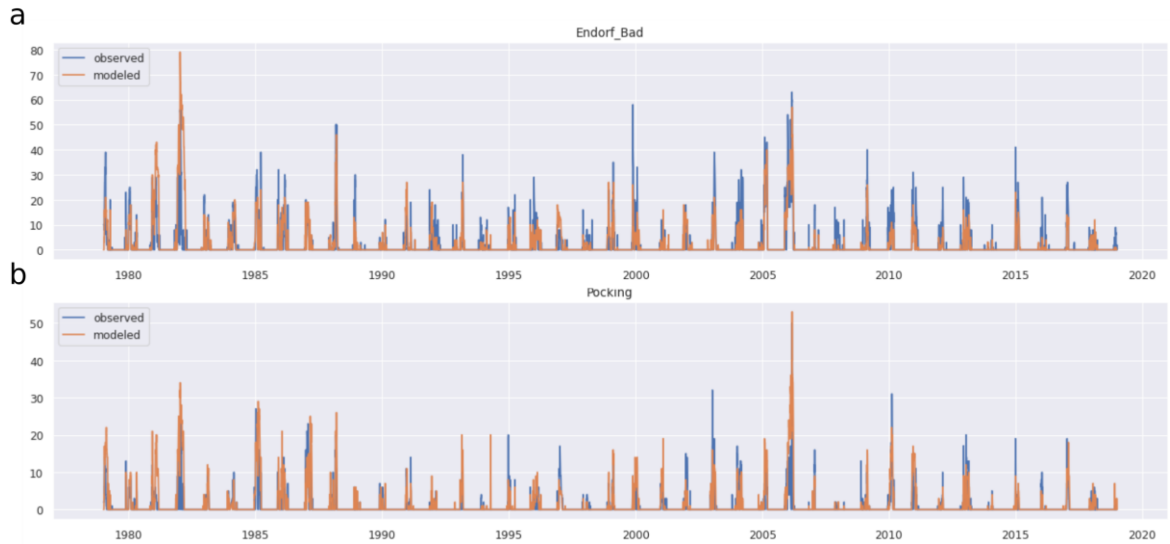

Figure 13: Stations with low differences in elevation relative to the SNOWGRID grid cell elevation show rather good agreement.

Too simple parametrization scheme

As already discussed in section "", some processes occur on scales smaller than the model grid resolution or are not captured by simple parametrization schemes (e.g. missing wind redistribution). This is especially true for glaciers which should be excluded from further analysis using the glacier mask (mask6), but might also apply to other locations such as mountain ridges etc., and especially contribute to error in mountainous areas.

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Conclusions

- very good agreement below 800m elevation
- differences above mainly caused by magnitude of modeled snow height, elevation and elevation delta
- differences due to bias (offset) which may be caused by:
    - biased input data
    - not modeled redistribution processes
    - scale effects
- even if biased, year to year variation is well covered
- thus, snow pack index is of much higher quality than absolute measures

Given the presence of scale effects and associated problems, as discussed in section “”, we refrain from publishing absolute snow depth or snow water equivalent data. Snow pack index is potentially more robust and less likely used in a deficient manner, as it is independent from absolute offsets.

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Supplementary figures

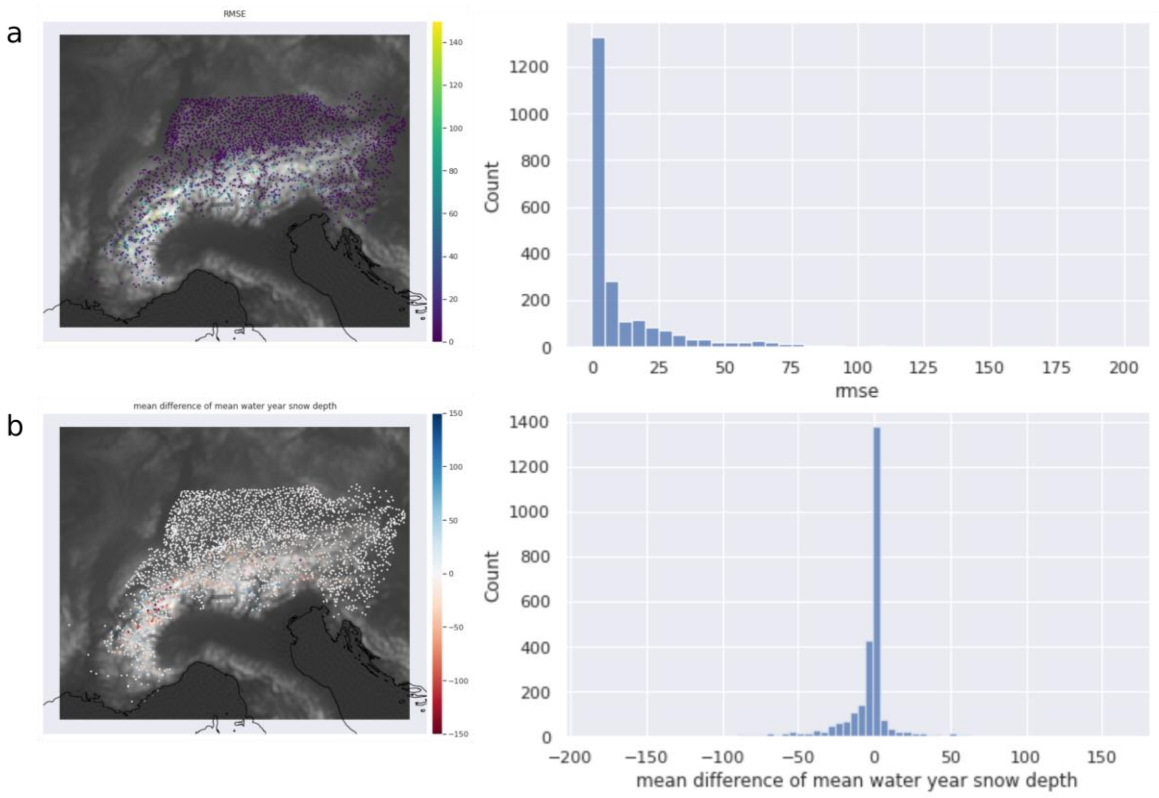

Figure 14: mean water year snow depth: a) RMSE, b) mean difference.

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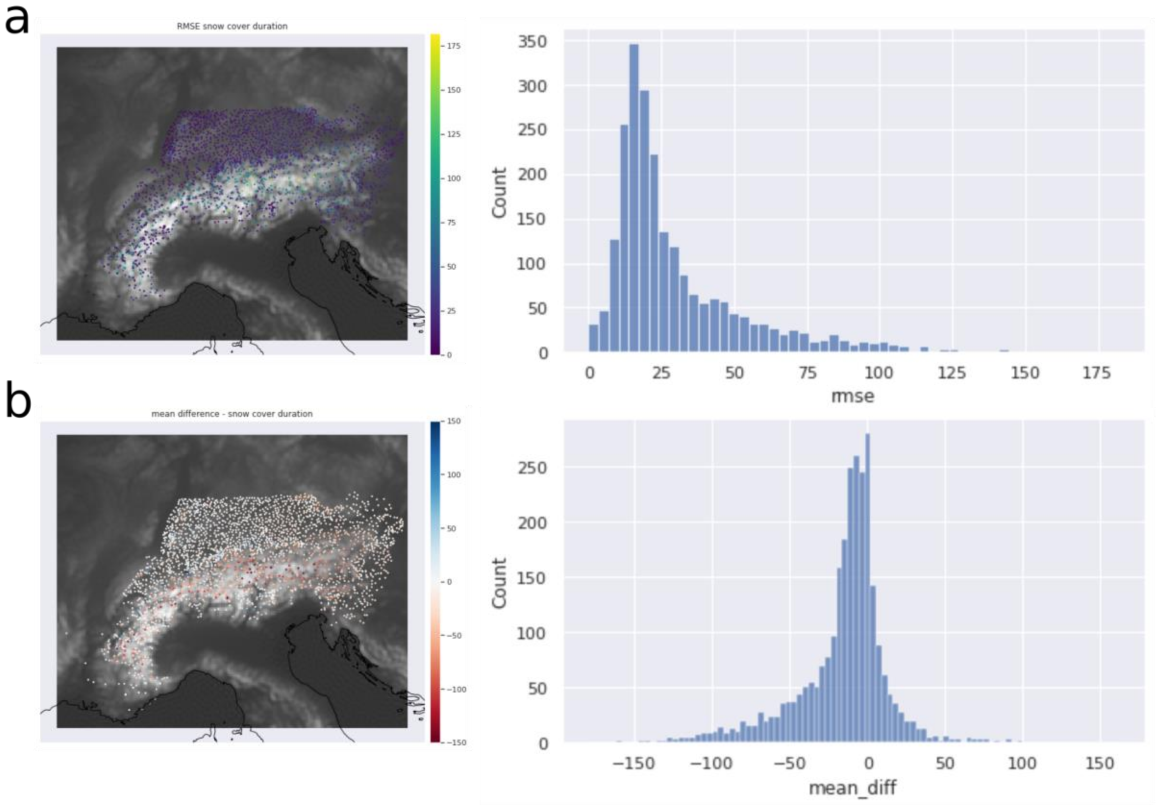

Figure 15: snow cover duration: a) RMSE, b) mean difference

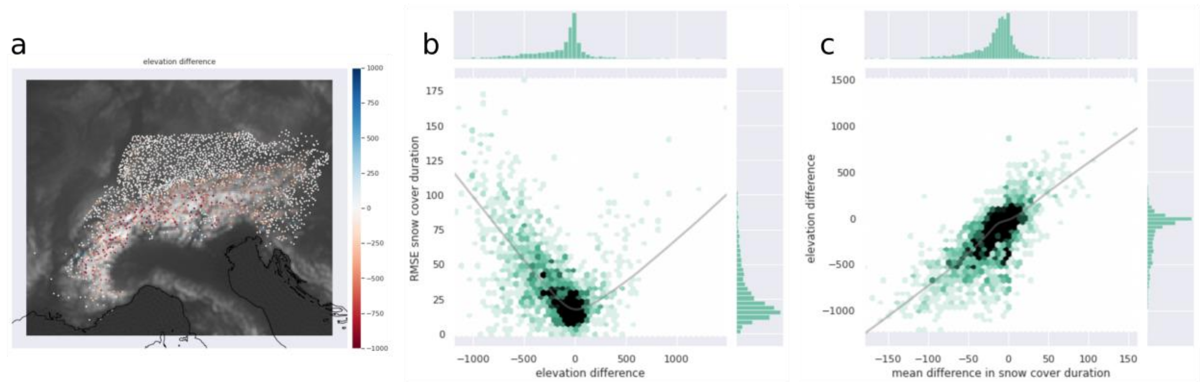

Figure 16: a) Difference in elevation between station and SNOWGRID grid cell (negative, red if station is lower; positive, blue if station is higher); b) snow cover duration RMSE gets higher at higher differences in elevation; c) negative difference (station lower) is associated with a negative difference in snow cover duration (higher modeled values), positive elevation differences with positive differences in snow cover duration (lower modeled values). Gray lines represent LOWESS smothers.

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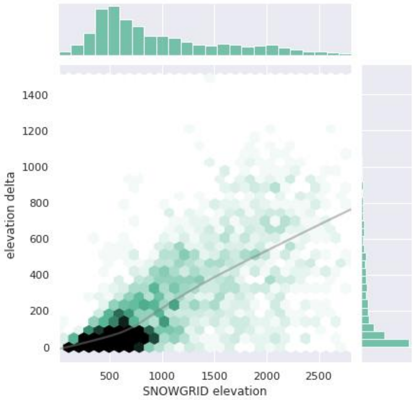

Figure 17:

elevation delta as a function of grid cell elevation

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